

Task 6 Password Strength Evaluation Report

Objective

To understand the characteristics of strong passwords and evaluate their security using an online password strength checker.

Tools Used

- **Windows (Google Chrome)**
- **PasswordMeter10**
- **Web Browser**

Password	Length	Complexity	Strength
password123	11	Lowercase + Numbers	Weak
Welcome2025	11	Uppercase + Lowercase + Numbers	Weak
W3lc0me@2025ddcfccddr	21	Uppercase + Lowercase + Numbers + Symbol	Very Strong
Xy#9Lp@4Qz!7Mv	15	Mixed Case + Numbers + Symbols	Very Strong

Evaluation Results

Password 1: password123

- Common dictionary word
- Predictable number sequence
- No special characters

Result: Weak password; vulnerable to dictionary attacks.

Password 2: Welcome2025

- Contains uppercase and lowercase letters
- Includes numbers
- Easy-to-guess word

Result: Medium strength.

Password 3: W3lc0me@2025

- Mixed case letters

- Numbers and symbols
- Increased complexity

Result: Strong password.

Password 4: Xy#9Lp@4Qz!7Mv

- 15 characters long
- Randomly generated
- Uses uppercase, lowercase, numbers, and symbols

Result: Very strong password.

Common Password Attacks

1. Brute Force Attack

An attacker tries every possible combination until the correct password is found.

Protection:

- Use long passwords (12+ characters)
- Include multiple character types

2. Dictionary Attack

An attacker uses a list of common words and passwords.

Protection:

- Avoid dictionary words
- Use random character combinations

3. Credential Stuffing

Attackers use leaked passwords from previous data breaches.

Protection:

- Use unique passwords for every account
- Enable Multi-Factor Authentication (MFA)

Password Security Best Practices

- Use at least 12–16 characters.
- Include uppercase and lowercase letters.
- Add numbers and special symbols.
- Avoid personal information.
- Do not reuse passwords.
- Use a password manager.
- Enable MFA.
- Change compromised passwords immediately.

Key Learnings

- Password length significantly increases security.
- Complexity makes passwords harder to guess.

- Randomly generated passwords provide the best protection.
- Strong passwords reduce the risk of brute-force and dictionary attacks.
- Password managers help create and store secure passwords.

Conclusion

Password strength depends on both length and complexity. Passwords containing uppercase letters, lowercase letters, numbers, and special characters are much more secure than simple passwords

Outputs

The image shows a password strength checker interface. At the top, the password 'password123' is entered in a dark input field. To the right of the input field are icons for a password strength indicator and a copy icon. Below the input field is a horizontal bar with five segments, the first of which is red, indicating a 'Very Weak' strength. To the right of the bar, the text 'Very Weak' is displayed in red. Below the bar, a red warning icon (a circle with an 'x') is followed by the text 'This is one of the most commonly used passwords'. Below this warning is a section titled 'TIME TO CRACK' with a subtitle '~undefined bits'. This section contains three rows of information: 1. 'Guessing online' with a description 'Trying passwords on a website — servers slow attackers to ~1,000/sec' and a result 'instantly' in red. 2. 'Leaked database — secure' with a description 'Password stored with modern encryption (bcrypt/scrypt) — ~10K guesses/sec' and a result 'instantly' in red. 3. 'Leaked database — fast' with a description 'Password stored with weak or no encryption (MD5/SHA-1) — ~100B guesses/sec' and a result 'instantly' in red. At the bottom of the interface, the word 'GENERATOR' is centered between two horizontal lines.

TIME TO CRACK		~undefined bits
Guessing online	Trying passwords on a website — servers slow attackers to ~1,000/sec	instantly
Leaked database — secure	Password stored with modern encryption (bcrypt/scrypt) — ~10K guesses/sec	instantly
Leaked database — fast	Password stored with weak or no encryption (MD5/SHA-1) — ~100B guesses/sec	instantly

GENERATOR

Welcome2025



Weak

⚠ 12+ characters is the recommended minimum

⚠ Contains a date or year — easily guessed

PATTERN MAP



■ Date pattern

TIME TO CRACK

~57 bits

Guessing online

Trying passwords on a website — servers slow attackers to
~1,000/sec

3M years

Leaked database — secure

Password stored with modern encryption (bcrypt/scrypt) —
~10K guesses/sec

392K years

Leaked database — fast

Password stored with weak or no encryption (MD5/SHA-1) —
~100B guesses/sec

9 days

GENERATOR

W3lc0me@2025ddcfccddr



Very Strong

⚠ Contains a date or year — easily guessed

☆ Excellent length

☆ Uses all four character types

PATTERN MAP



■ Date pattern

TIME TO CRACK

~130 bits

Guessing online

Trying passwords on a website — servers slow attackers to ~1,000/sec

> 1 trillion years

Leaked database — secure

Password stored with modern encryption (bcrypt/scrypt) — ~10K guesses/sec

> 1 trillion years

Leaked database — fast

Password stored with weak or no encryption (MD5/SHA-1) — ~100B guesses/sec

> 1 trillion years

GENERATOR

Xy#9Lp@4Qz!7Mv



Strong

- ✓ Good length
- ☆ Uses all four character types
- ☆ No common patterns detected

TIME TO CRACK

~92 bits

Guessing online

Trying passwords on a website — servers slow
attackers to ~1,000/sec

> 1 trillion years

Leaked database — secure

Password stored with modern encryption
(bcrypt/scrypt) — ~10K guesses/sec

> 1 trillion years

Leaked database — fast

Password stored with weak or no encryption (MD5/SHA-1)
— ~100B guesses/sec

484M years

GENERATOR